S.M. MAHBUBUL HASAN

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SUMMARY

CSE graduate with strong practical experience in Python, HTML, and CSS, and solid expertise in Deep Learning and Computer Vision. Built and deployed real-world AI projects including waste classification using YOLOv8 and brain tumor segmentation using SegFormer. Skilled in developing end-to-end machine learning pipelines and integrating AI models with web-based applications. Eager to contribute to building scalable and impactful AI solutions.

PROJECT

Real-Time Waste Classification System Using YOLOv8, Senior Design Project

North South University

- Trained a YOLOv8s object detection model on 11,709 images across 23 waste categories, achieving mAP@0.5 of 0.919, precision of 0.904, and recall of 0.882
- Built and deployed a real-time web application using Streamlit & OpenCV supporting live webcam inference, batch image processing, and automated waste sorting simulation
- Applied data augmentation, normalization, and overfitting mitigation strategies (early stopping, dropout, weight decay) to maximize model generalization
- Tech stack: Python · PyTorch · YOLOv8 · OpenCV · Streamlit · Roboflow · Google Colab (NVIDIA Tesla T4)

Brain Tumor Segmentation (SegFormer),Directed Research

North South University

- Developed a deep learning model using SegFormer for brain tumor segmentation in medical images
- Applied preprocessing techniques (resizing, normalization, augmentation) to enhance model performance
- Trained and evaluated model for pixel-level tumor detection and segmentation
- Tech Stack: Python, PyTorch, SegFormer, OpenCV, Google Colab

COURSEWORK

CSE 445: Machine Learning

North South University Department of Electrical and Computer Engineering School of Engineering & Physical Sciences • 2025 • Hands-on understanding of supervised & unsupervised learning, neural networks, clustering techniques, decision trees, and evolutionary algorithms.

CSE 440: Artificial Intelligence

North South University Department of Electrical and Computer Engineering School of Engineering & Physical Science • 2025 • Gained knowledge in AI techniques including search/problem solving, knowledge representation, machine learning, NLP, computer vision, and robotics with exposure to LISP & PROLOG.

CSE 273: Introduction to Theory of Computation

North South University Department of Electrical and Computer Engineering School of Engineering & Physical Sciences • 2024 • Theorem proving, propositional logic, first order logic, finite automata, formal languages, Turing machines, uncomputability, computational complexity and NP completeness.

SKILLS

Programming Languages: Python, C, C++, Java
Web Technologies: HTML, CSS
Machine Learning & AI: Deep Learning, Computer Vision (YOLOv8, SegFormer, EfficientNetB4)
Tools & Platforms: Git & GitHub, TensorFlow / PyTorch, OpenCV, Streamlit, Trello
Core Concepts: Data Preprocessing & Augmentation, Model Training & Evaluation, Image Classification & Segmentation

EDUCATION

Bachelor of Science in Computer Science and Engineering

NORTH SOUTH UNIVERSITY • Bashundhara, Dhaka-1229, Bangladesh • 2025